

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

A bookstore manages its inventory, customers, and sales. The main entities are 'Book', which has attributes like title, author, and ISBN; 'Customer', which has attributes like customerID, name, and contactNumber; and 'Sale', which records interactions between customers and books with attributes such as saleID, saleDate, and quantity. The relationships include 'purchases' where a customer can purchase multiple books and each book can be purchased by many customers.

- a) Identify the entities and their attributes in the given scenario. (5)
- b) Specify the relationships and their cardinalities between the identified entities. (5)
- c) Draw an ER diagram representing the entities, their attributes, and relationships. (5)
- d) Map the ER diagram to a relational schema, including appropriate primary and foreign keys. (5)

Question Q2 (15 marks)

Consider a database management system scenario.

- a) Define the key concepts and their importance related to frame bytes time. (3)
- b) Describe the main components and how they interact related to frame bytes time. (1)
- c) Analyze the advantages and disadvantages of different approaches related to frame bytes time. (2)
- d) Compare different implementation strategies and their trade-offs related to frame bytes time. (2)
- e) Evaluate the effectiveness of the proposed solution related to frame bytes time. (2)
- f) Explain the real-world applications and use cases related to frame bytes time. (2)
- g) List the key features and their benefits related to frame bytes time. (3)

Question Q3 (25 marks)

Consider a database management system scenario.

- a) Define the key concepts and their importance related to Transaction Management and Concurrency. (4)
- b) Describe the main components and how they interact related to Transaction Management and Concurrency. (2)
- c) Analyze the advantages and disadvantages of different approaches related to Transaction Management and Concurrency. (3)
- d) Compare different implementation strategies and their trade-offs related to Transaction Management and Concurrency. (5)
- e) Evaluate the effectiveness of the proposed solution related to Transaction Management and Concurrency. (5)
- f) Explain the real-world applications and use cases related to Transaction Management and Concurrency. (3)
- g) List the key features and their benefits related to Transaction Management and Concurrency. (3)

Question Q4 (40 marks)

In a local bookstore, three primary entities are involved: Books, Authors, and Publishers. Each Book has attributes such as ISBN, title, and price. Authors have names, birthdates, and nationality, while Publishers encompass a name and address. A Book can have multiple Authors and is published by one Publisher. An Author can write multiple Books, and a Publisher can publish multiple Books. Define several aspects of this scenario as you analyze its database structure.

- a) Identify the primary entities in the bookstore scenario and list their key attributes. (10)
- b) Analyze the relationships between Books, Authors, and Publishers, indicating their cardinalities. (10)
- c) Describe the specialization and generalization concepts that might apply to the Authors entity, considering possible subtypes like 'Fiction' and 'Non-Fiction' writers. (10)
- d) Discuss a potential ternary relationship in this scenario and explain its significance in capturing shared attributes among the entities. (10)